Stat 266 – Unit 9 – Video Set 2 – Class Examples

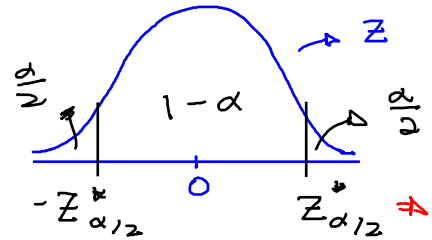
Summary: Related to the Normal distribution.

(*Z-tools*) Confidence Intervals

1. If we have an estimate such that $\boxed{\hat{\theta} \sim Normal(\mu_{\hat{\theta}} = \theta, \sigma^2_{\hat{\theta}})}$, we have the following pivotal quantity,

$$Z = \frac{\hat{\theta} - \mu_{\hat{\theta}}}{\sigma_{\hat{\theta}}} = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim \underbrace{Normal(0,1)}_{\text{doesn't depend on } \theta}.$$

standardize



critical value

2. For $Z \sim Normal(0,1)$, $P(-z^*_{\alpha/2} \le Z \le z^*_{\alpha/2}) = 1 - \alpha$.

3. Then, $P(-z^*_{\alpha/2} \le Z \le z^*_{\alpha/2}) = P\left(-z^*_{\alpha/2} \le \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \le z^*_{\alpha/2}\right) = P(\hat{\theta} - z^*_{\alpha/2}\sigma_{\hat{\theta}} \le \theta \le \hat{\theta} + z^*_{\alpha/2}\sigma_{\hat{\theta}}) = 1 - \alpha.$

↳ if $\sigma_{\hat{\theta}}$ is NOT a function of θ.

$\Rightarrow$ A two-sided $Z - Tools$ $(1-\alpha)100\%$ confidence interval for θ is $\hat{\theta} \pm \underbrace{z^*_{\alpha/2}\sigma_{\hat{\theta}}}_{\text{margin of error.}}$.

Note: If $\sigma_{\hat{\theta}}$ is a function of θ, solve $P\left(-z^*_{\alpha/2} \le \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \le z^*_{\alpha/2}\right)$ for $P(\hat{\theta}_L \le \theta \le \hat{\theta}_U)$ accordingly.

Note: If $\hat{\theta} \approx Normal(\mu_{\hat{\theta}} = \theta, \sigma^2_{\hat{\theta}})$ for large sample sizes then $\hat{\theta} \pm z^*_{\alpha/2}\sigma_{\hat{\theta}}$ is an approximate (large sample) $Z - Tools$ confidence interval for θ.

Common Applications:

1. Linear Combinations of Means.
2. Linear Combinations of Proportions.

Example A : Suppose $\{Y_1,\ldots,Y_n\}$ is a random sample of independent and identically distributed $Uniform(0,\theta)$ random variables.

Recall: An exact $(1-\alpha)100\%$ CI for θ is

$$\left(\frac{Y_{(n)}}{(1-\alpha/2)^{1/n}},\ \frac{Y_{(n)}}{(\alpha/2)^{1/n}}\right).$$

$$\Rightarrow f_Y(y)=\begin{cases}\dfrac{1}{\theta} & ,\ 0<y<\theta\\ 0 & ,\ else\end{cases}$$

Define an approximate Z-tools confidence interval for θ as a function of $\bar{Y}$.

From CLT, $\bar{Y} \approx Normal\left(\mu_{\bar{Y}} = \mu_Y = \frac{\theta}{2},\ \sigma^2_{\bar{Y}} = \frac{\sigma^2_Y}{n} = \frac{\theta^2}{12n}\right)$.

Parameter: $\theta = 2\mu_Y$

Estimate: $\hat{\theta} = 2\hat{\mu}_Y = 2\bar{Y} \approx Normal\left(\mu_{\hat{\theta}} = \mu_{2\bar{Y}} = \theta,\ \sigma^2_{\hat{\theta}} = \sigma^2_{2\bar{Y}} = 4\sigma^2_{\bar{Y}} = \frac{\theta^2}{3n}\right)$

1) Pivot: $$Z = \frac{\hat{\theta} - \mu_{\hat{\theta}}}{\sigma_{\hat{\theta}}} = \underbrace{\frac{2\bar{Y} - \theta}{\theta/\sqrt{3n}}}_{\sigma_{\hat{\theta}} \text{ depends on } \theta} = \frac{2\sqrt{3n}}{\theta}\bar{Y} - \sqrt{3n} \approx Normal(0,1).$$

$$1-\alpha = P\left(-Z^*_{\alpha/2} < Z < Z^*_{\alpha/2}\right) = P\left(-Z^*_{\alpha/2} < \frac{2\sqrt{3n}}{\theta}\bar{Y} - \sqrt{3n} < Z^*_{\alpha/2}\right)$$

$$= P\left(\underbrace{\frac{2\sqrt{3n}\cdot\bar{Y}}{\sqrt{3n} + Z^*_{\alpha/2}}}_{\hat{\theta}_L} < \theta < \underbrace{\frac{2\sqrt{3n}\cdot\bar{Y}}{\sqrt{3n} - Z^*_{\alpha/2}}}_{\hat{\theta}_U}\right),$$

2) Or, suppose we used $\hat{\Theta} \pm Z^*_{\alpha/2} \cdot \sigma_{\hat{\Theta}}$:

$$\Rightarrow 2\bar{Y} \pm Z^*_{\alpha/2} \cdot \frac{\Theta}{\sqrt{3n}}$$

- depends on Θ, which is unknown.
- is NOT a CI.

Since $\hat{\Theta} = 2\bar{Y}$,

$$\Rightarrow 2\bar{Y} \pm Z^*_{\alpha/2} \cdot \frac{2\bar{Y}}{\sqrt{3n}} \Rightarrow \left(2\bar{Y}\left(1 - \frac{Z^*_{\alpha/2}}{\sqrt{3n}}\right), 2\bar{Y}\left(1 + \frac{Z^*_{\alpha/2}}{\sqrt{3n}}\right) \right)$$

Example B : Suppose $\{Y_1,\ldots,Y_n\}$ is a random sample of independent and identically distributed $Exponential(\theta)$ random variables.

Parameter : $\theta = \mu_Y$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , y > 0 \\ 0 & , else \end{cases}, \quad \theta > 0$$

An exact C.I. for θ

$$\left(\frac{2n\bar{Y}}{\chi^{2*}_{2n,\alpha/2}} , \frac{2n\bar{Y}}{\chi^{2*}_{2n,1-\alpha/2}} \right).$$

Define an approximate Z-tools confidence interval for θ as a function of $\bar{Y}$.

From CLT, $\bar{Y} \approx Normal\left(\mu_{\bar{Y}} = \mu_Y = \theta , \underbrace{\sigma^2_{\bar{Y}} = \frac{\sigma^2_Y}{n} = \frac{\theta^2}{n}}_{\text{depends on } \theta}\right)$

Pivot: $Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} = \frac{\bar{Y} - \theta}{\theta/\sqrt{n}} = \frac{\bar{Y}\sqrt{n}}{\theta} - \sqrt{n} \approx N(0,1)$, for large n.

1) $1 - \alpha = P\left(-Z^*_{\alpha/2} < \frac{\bar{Y}\sqrt{n}}{\theta} - \sqrt{n} < Z^*_{\alpha/2}\right)$

$$= P\left(\underbrace{\frac{\bar{Y}\sqrt{n}}{\sqrt{n} + Z^*_{\alpha/2}}}_{\hat{\theta}_L} < \theta < \underbrace{\frac{\bar{Y}\sqrt{n}}{\sqrt{n} - Z^*_{\alpha/2}}}_{\hat{\theta}_U} \right)$$

2) Using $\hat{\theta} \pm Z^*_{\alpha/2} \sigma_{\hat{\theta}}$

$$\Rightarrow \bar{Y} \pm Z^*_{\alpha/2} \frac{\theta}{\sqrt{n}} \approx \bar{Y} \pm Z^*_{\alpha/2} \cdot \frac{\bar{Y}}{\sqrt{n}} \Rightarrow \left(\bar{Y}\left(1 - \frac{Z^*_{\alpha/2}}{\sqrt{n}}\right), \bar{Y}\left(1 + \frac{Z^*_{\alpha/2}}{\sqrt{n}}\right) \right)$$

Suppose $n = 50$ and we observed $\bar{y} = 10$. Consider 95% CI's for $\mu_Y = \theta$.

1) (Exact): $\left(\dfrac{2n\bar{y}}{\chi^2_{2n,\alpha/2}} , \dfrac{2n\bar{y}}{\chi^2_{2n,1-\alpha/2}} \right) \Rightarrow \left(\dfrac{2(50)(10)}{129.561} , \dfrac{2(50)(10)}{74.222} \right)$

$\Rightarrow (7.7184, 13.4731)$

2) (Approx 1): $\left(\dfrac{\bar{y}\sqrt{n}}{\sqrt{n} + z^*_{\alpha/2}} , \dfrac{\bar{y}\sqrt{n}}{\sqrt{n} - z^*_{\alpha/2}} \right) \Rightarrow \left(\dfrac{10\sqrt{50}}{\sqrt{50} + 1.96} , \dfrac{10\sqrt{50}}{\sqrt{50} - 1.96} \right)$

$\Rightarrow (7.8297, 13.8348)$

3) (Approx 2): $\left(\bar{y}\left(1 - \dfrac{z^*_{\alpha/2}}{\sqrt{n}}\right) , \bar{y}\left(1 + \dfrac{z^*_{\alpha/2}}{\sqrt{n}}\right) \right)$

$\Rightarrow \left(10\left(1 - \dfrac{1.96}{\sqrt{50}}\right) , 10\left(1 + \dfrac{1.96}{\sqrt{50}}\right) \right) \Rightarrow (7.2281, 12.7719)$

Example C : Suppose $\{Y_1,\ldots,Y_n\}$ is a random sample of independent and identically distributed random variables with density:

$$\Rightarrow f_Y(y)=\begin{cases} e^{-(y-\theta)} & , \quad y>\theta \\ 0 & , \quad \text{else} \end{cases}$$

$\Rightarrow Y-\theta \sim Exp(1)$

$\mu_Y = \theta + 1 \quad , \quad \sigma_Y^2 = 1$

$\Rightarrow \theta = \mu_Y - 1$

Define an approximate Z-tools confidence interval for θ as a function of $\bar{Y}$.

- For large n, $\bar{Y} \approx Normal(\mu_{\bar{Y}} = \mu_Y = \theta + 1, \ \sigma_{\bar{Y}}^2 = \sigma_Y^2/n = 1/n)$

- $\hat{\theta} = \bar{Y} - 1 \approx Normal(\mu_{\hat{\theta}} = \theta, \ \sigma_{\hat{\theta}}^2 = 1/n)$

- $Z = \dfrac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} = \dfrac{(\bar{Y}-1) - \theta}{1/\sqrt{n}} = (\bar{Y}-1)\sqrt{n} - \theta\sqrt{n} \approx Normal(0,1).$

- $1-\alpha = P\left(-Z^*_{\alpha/2} < (\bar{Y}-1)\sqrt{n} - \theta\sqrt{n} < Z^*_{\alpha/2}\right)$

$$= P\Big(\underbrace{\bar{Y} - 1 - \frac{1}{\sqrt{n}} Z^*_{\alpha/2}}_{\hat{\theta}_L} < \theta < \underbrace{\bar{Y} - 1 + \frac{1}{\sqrt{n}} Z^*_{\alpha/2}}_{\hat{\theta}_U}\Big)$$

Example D : Suppose you have two independent samples from different populations as follows:

- n_1 observations from $X \sim Normal(\mu_X, \sigma_X^2)$ and
- n_2 observations from $Y \sim Normal(\mu_Y, \sigma_Y^2)$.

Define a confidence interval for $\mu_X - \mu_Y$.

- The MVUE of a Normal is the sample mean.
- $\bar{X}$ is the MVUE for μ_X,

  $\bar{X} \sim Normal(\mu_{\bar{X}} = \mu_X, \sigma_{\bar{X}}^2 = \sigma_X^2/n_1)$
- $\bar{Y}$ is the MVUE for μ_Y,

  $\bar{Y} \sim Normal(\mu_{\bar{Y}} = \mu_Y, \sigma_{\bar{Y}}^2 = \sigma_Y^2/n_2)$

Aside: For one mean, μ_X, a $(1-\alpha)100\%$ CI is

$$\hat{\theta} \pm Z^*_{\alpha/2}\, \sigma_{\hat{\theta}} \Rightarrow \bar{x} \pm \left\{ Z^*_{\alpha/2} \cdot \frac{\sigma_X}{\sqrt{n_1}} \right\}$$

Note: σ_X^2 is unknown. The MVUE for σ_X^2 (when X is Normal) is $S^2 \Rightarrow$ the sample variance.

For large n, an approximate $(1-\alpha)100\%$ CI for μ_Y is $\Rightarrow \bar{x} \pm \left\{ Z^*_{\alpha/2} \cdot \frac{S_x}{\sqrt{n_1}} \right\}$.

Parameter: $\mu_X - \mu_Y$

Estimate: $\bar{X} - \bar{Y}$ (a linear combination of Normals)

$$\hat{\Theta} = \bar{X} - \bar{Y} \sim \text{Normal}\left(\mu_{\hat{\Theta}} = \mu_{\bar{X}-\bar{Y}} = E[\bar{X}-\bar{Y}] = E[\bar{X}] - E[\bar{Y}] = \mu_X - \mu_Y,\right.$$

$$\left.\sigma^2_{\hat{\Theta}} = V[\bar{X}-\bar{Y}] = V[\bar{X}] + V[\bar{Y}] = \frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}\right)$$

independent

$$\Rightarrow \hat{\Theta} \pm z^*_{\alpha/2}\,\sigma_{\hat{\Theta}} = (\bar{X}-\bar{Y}) \pm z^*_{\alpha/2}\underbrace{\sqrt{\frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}}}$$

We would need to know the values of σ_X^2 and σ_Y^2, or estimate them.

For large n_1 and n_2, an approximate CI for $\mu_X - \mu_Y$,

$$(\bar{X}-\bar{Y}) \pm z^*_{\alpha/2}\sqrt{\frac{S_X^2}{n_1} + \frac{S_Y^2}{n_2}}.$$

Example E: An experiment to explore the pain thresholds to electrical shocks for males and females resulted in the data summary below. Suppose the pain thresholds for both males and females are normally distributed with known variance stated in the table below.

	Males	Females
n	$n_1 = 14$	$n_2 = 10$
Sample mean	$\bar{x} = 16.2$	$\bar{y} = 14.9$
Population Variance	$\sigma_x^2 = 12.7$	$\sigma_y^2 = 26.4$

Define a (1 - α)100% confidence interval for the difference in population means between males and females.

Consider 95% CI's: $z^*_{\alpha/2} = z^*_{0.025} = 1.96$

Males (μ_x): $\bar{x} \pm z^*_{\alpha/2} \cdot \frac{\sigma_x}{\sqrt{n_1}} \Rightarrow 16.2 \pm 1.96\left(\frac{\sqrt{12.7}}{\sqrt{14}}\right) \Rightarrow (14.333, 18.067)$

Females (μ_y): $14.9 \pm 1.96\left(\frac{\sqrt{26.4}}{\sqrt{10}}\right) \Rightarrow (11.715, 18.085)$.

Is there a real difference between μ_x and μ_y?

$\theta = \mu_x - \mu_y$, $\hat{\theta} = \bar{x} - \bar{y} = 16.2 - 14.9 = 1.30$,

$\sigma_{\hat{\theta}} = \sqrt{\frac{\sigma_x^2}{n_1} + \frac{\sigma_y^2}{n_2}} = \sqrt{\frac{12.7}{14} + \frac{26.4}{10}} = 1.8834$.

A 95% CI for $\mu_x - \mu_y$ is then

$1.30 \pm \{1.96(1.8834)\} \Rightarrow 1.30 \pm 3.6915 \Rightarrow (-2.39, 4.99)$.

Linear Combinations of Means: Consider k populations $Y_1, Y_2, \ldots, Y_k$ where

$$Y_i \overset{ind}{\sim} \text{Normal}(\mu_{Y_i}, \sigma^2_{Y_i}), \quad i = 1, \ldots, k.$$

Parameter: $\theta = c_1 \mu_{Y_1} + c_2 \mu_{Y_2} + \cdots + c_k \mu_{Y_k}$, $c_1, \ldots, c_k$ are chosen constants

Estimate: $\hat{\theta} = \underbrace{c_1 \bar{Y}_1 + c_2 \bar{Y}_2 + \cdots + c_k \bar{Y}_k}$

- a linear combination of sample means (all Normal)
- a linear combination of Normals

$$\hat{\theta} \sim \text{Normal}\left(\mu_{\hat{\theta}} = E[\hat{\theta}] = \theta = c_1 \mu_{Y_1} + \cdots + c_k \mu_{Y_k}, \right.$$

$$\left. \sigma^2_{\hat{\theta}} = V[\hat{\theta}] = c_1^2 \cdot \frac{\sigma^2_{Y_1}}{n_1} + c_2^2 \cdot \frac{\sigma^2_{Y_2}}{n_2} + \cdots + c_k^2 \cdot \frac{\sigma^2_{Y_k}}{n_k} \right)$$

So $\hat{\theta} \pm Z^*_{\alpha/2} \sigma_{\hat{\theta}}$

$$\Rightarrow \sum_{i=1}^{k} c_i \bar{Y}_i \pm Z^*_{\alpha/2} \sqrt{\sum_{i=1}^{k} \frac{c_i^2 \sigma^2_{Y_i}}{n_i}}.$$

<u>Note</u>: For large n's, $S^2_{Y_i} \approx \sigma^2_{Y_i}$.

Example F: A random sample of homeowners and condominium owners were asked whether they planned any landscaping within the next year. The data are given in the following table.

Residence	Sample size	Number who are Planning to landscape
Homeowner	260	91
Condominium owner	300	60

$\hat{p}_1 = 91/260 = 0.35$

$\hat{p}_2 = 60/300 = 0.20$

Construct a 90% confidence interval for $p_1 - p_2$, where p_1 and p_2 are the population proportion of people who are planning to landscape for homeowners and condominium owner, respectively.

From CLT, $\hat{p} \approx \text{Normal}\left(\mu_{\hat{p}} = p,\ \sigma^2_{\hat{p}} = \frac{p(1-p)}{n}\right)$.

An approximate $(1-\alpha)100\%$ CI for p

$$\approx \hat{p} \pm Z^*_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

$\theta = p_1 - p_2$, $\hat{\theta} = \hat{p}_1 - \hat{p}_2 \approx \text{Normal}\left(\mu_{\hat{\theta}} = p_1 - p_2,\ \sigma^2_{\hat{\theta}} = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$

An approximate $(1-\alpha)100\%$ CI for $p_1 - p_2$ is

$$(\hat{p}_1 - \hat{p}_2) \pm Z^*_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}.$$

$$\Rightarrow (0.35 - 0.20) \pm 1.645\sqrt{\frac{0.35(0.65)}{260} + \frac{0.20(0.80)}{300}} \Rightarrow 0.15 \pm 0.06173$$

$$\Rightarrow (0.0883, 0.2117)$$